

PRANAV GOYAL

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EDUCATION

University of Michigan, Ann Arbor

BSE Computer Engineering, Minor in Entrepreneurship

May 2027

GPA: 3.6/4.00, Fall 2023, Winter 2024 & Fall 2025 Dean's Honor List

Coursework: Data Structures and Algorithms, Computer Architecture, Embedded Systems, Computer Networks, Operating Systems

WORK EXPERIENCE

Commvault

Software Engineering Intern

May 2025 - August 2025

- Built a semi-supervised ML model to detect potential security threats in data backups for enterprise data solutions using feedforward neural network built using TensorFlow and Keras, achieving 95% accuracy in detecting anomalous backups and ransomware
- Wrote advanced PowerShell and SQL scripts to automate collection of data from production environments to limit downtime and created a data labeling pipeline using an ensemble of unsupervised anomaly detection models including Isolation Forest models
- Implemented LLM based techniques to enhance proactive detection capabilities in existing models, integrating with C++ codebase

Microsoft

Software Developer Intern

June 2024 - August 2024

- Developed a "Shop the Look" omnichannel solution with a chatbot and robot for physical stores, enabling natural language, voice and image-based searches in fashion retail catalogs to enhance and personalize traditional apparel search experiences
- Built LLM Agents and implemented GenAI and multimodal search using Python and Azure OpenAI APIs, projected to boost conversion rates and customer engagement by 20% in online and physical apparel stores
- Engineered RAG and vector search pipelines using Azure Search and OpenAI Assistants, improving product discovery by 30%
- Utilized Agile, CI/CD pipelines and DevOps best practices to deliver production ready, well-tested application within six weeks and deployed on Docker for Microsoft customer demos

PROJECTS

Michigan Embedded Systems Security Team (MITRE eCTF) | C, Python

- Developed secure firmware for Cortex-M0+ based MCU, implementing cryptographic protections against side-channel attacks
- Engineered automated agentic fuzzing pipeline, orchestrated by LLM agents, to identify vulnerabilities in rival teams' designs, utilizing QEMU to virtualize microcontroller and peripherals, and a suite of fuzzers to simulate malicious inputs

Perot Jain Tech Lab at Mcity, Spin Mobility | Python

- Developed unsupervised ML model to identify hazardous micro-mobility zones by analyzing GPS and IMU telemetry for anomalies
- Engineered a V2I communication system linking traffic signals and scooters to optimize traffic flow and implement priority signaling

Reliable Transport Protocol | C++

- Built a Reliable Transport Protocol over UDP, featuring a sliding window flow control, retransmit, and congestion control algorithms
- Validated protocol stability by simulating high-latency, 10% packet-loss network environments, achieving 100% data integrity

Autonomous Vehicle Systems & ADAS Firmware | C

- Engineered autonomous vehicle prototype using STM32 firmware and custom CAD-designed chassis; implemented PID for lane centering and sensor-fusion logic with ultrasonic and reflectance sensors for adaptive cruise control and automated lane changes
- Wrote a custom SPI-to-XBEE communication protocol to facilitate bidirectional wireless control across a multi-vehicle fleet

RISC Processor Prototype | Verilog, C++

- Architected efficient superscalar, OoO processor for custom RISC ISA, implementing data forwarding to maintain data integrity
- Designed and built a C++ toolchain (compiler, assembler, linker) that performs full assembly-to-binary translation, including robust label resolution, enabling complex branching, and engineered a simulator to model CPU state, enabling end-to-end verification

CaseCraft – AI Lawyer | Python, JavaScript (Next.js)

- Engineered legal RAG pipeline using Gemini grounded in 22 M words of US code, achieving 80% accuracy in precise legal guidance

LEADERSHIP

Tau Epsilon Kappa Professional Technology Fraternity

Vice President, Professional Development

May 2025 - Present

- Organized and conducted networking events with top technology companies, boosting internship offer rates for members by 10%
- Introduced projects for members to gain real-life software engineering skills through a semester-long project with industry partner

Shift Creator Space

Executive Director

Jan 2026 - Present

- Raised \$5,000+ in funding for member projects, via partnerships with various on-campus resources and led redesign of shared space
- Implemented IoT automation with Slack integration to allow real-time updates on space usage, and encourage member attendance

SKILLS AND ACTIVITIES

Programming Languages: C++, C, Python, MATLAB, HTML, CSS, JavaScript (React, Next.js, Node.js), SV/Verilog, Tcl

Technologies: git, Docker, LLMs (OpenAI, Gemini), Azure, Linux, Onshape, Microsoft Office, Figma

Activities: General Member of MRacing – Powertrain division

Interests: Wildlife and bird photography, Music and drumming (10+ years of experience), Chess (1300 Online Rating)